

rag-quality-evaluator: faithfulness, citation grounding, and drift detection for RAG systems

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1 Abstract

We present rag-quality-evaluator, a multi-metric quality evaluator for RAG systems with built-in drift detection across runs. The package implements six RAG-specific metrics (faithfulness,

answer relevance, citation grounding, context precision, context recall, answer correctness) with two judge backends (a keyless heuristic for CI and an LLM-as-judge for headline numbers). The drift detector runs a Welch’s t-test between two run snapshots and flags any metric that moved by ≥ 0.03 absolute or with $p < 0.05$. We demonstrate the harness on a 10-sample synthetic fixture with a deliberately-degraded candidate run and show the drift detector correctly flags the regression in `answer_correctness` (delta = -0.107, $p = 0.033$) while leaving the unrelated retrieval metrics untouched.

2 1. Background

RAG systems fail in several distinct ways that a single accuracy number hides:

- **Hallucination**: the answer makes factual claims the retrieved context does not support.
- **Citation grounding errors**: the answer cites the wrong chunks.
- **Retrieval failures**: the right context wasn’t retrieved.
- **Over-summarization or under-summarization**: the answer addresses only part of the question.

A practical RAG quality evaluator therefore reports a *set* of metrics, each targeting a specific failure mode. RAGAS (Es et al., 2023) is the standard implementation; this project follows its metric definitions but adds a keyless heuristic judge so the suite runs in CI, plus a drift-detection layer so the *change* across iterations becomes a first- class artifact.

3 2. Related Work

RAGAS (Es et al., 2023): the reference RAG eval framework. Our metric definitions match theirs.

Judging LLMs as judges. Zheng et al. (2023). We use the per-metric mini-rubric pattern rather than one omnibus prompt.

SelfCheckGPT (Manakul et al., 2023): hallucination detection by self-consistency. Not yet wired in but the API surface matches.

Drift detection. Standard two-sample t-tests on per-sample scores; nothing exotic, but treating drift detection as a first-class output of the eval pipeline is the contribution here.

4 3. Method

4.1 3.1 Metrics

metric	what it measures	judge
faithfulness	does the answer’s claims appear in the retrieved context?	both
answer_relevance	does the answer address the question?	both

metric	what it measures	judge
citation_grounding	if the answer cites chunks, do those chunks back it?	both
context_precision	fraction of retrieved chunks that are actually relevant	heur
context_recall	fraction of gold-answer terms covered by chunks	heur
answer_correctness	how close is the answer to the gold (cosine + overlap)	heur

4.2 3.2 Heuristic judge

Sentence-transformers (BGE-small) embeddings + token overlap (Jaccard). The metrics are coarser than the LLM judge but the relative ordering across samples is preserved, which is what the drift detector cares about.

4.3 3.3 LLM judge

Per-metric mini-rubric prompts: each judge call asks for one 0-1 score plus a one-line rationale, returned as JSON. Smaller rubrics produce more consistent JSON outputs at lower judge cost than one omnibus prompt. Council mode (N judges) is implemented.

4.4 3.4 Drift detection

```
from scipy.stats import ttest_ind
_, p = ttest_ind(baseline_scores, candidate_scores, equal_var=False)
delta = mean(candidate) - mean(baseline)
flagged = abs(delta) >= 0.03 OR p < 0.05
```

Welch's t-test (unequal variances) per metric. No multi-test correction in v1; that's the obvious next addition.

5 4. Data

The in-repo synthetic fixture has 10 hand-curated Q/A items. Two versions (`synthetic.jsonl` and `synthetic_v2.jsonl`); v2 has five intentionally-degraded answers (off-topic, vague, factually wrong) so the drift detector has something to find.

6 5. Evaluation Setup

Baseline + candidate runs through the heuristic judge, drift check, then five-chart plot generation. Hardware: Apple M-series CPU.

7 6. Results

Baseline run (`synthetic.jsonl`, $n=10$):

metric	mean	min	max
faithfulness	0.912	0.864	0.992
citation_grounding	0.909	0.864	0.992
context_recall	0.854	0.375	1.000
answer_relevance	0.811	0.681	0.875
context_precision	0.550	0.000	1.000
answer_correctness	0.535	0.337	0.651

Drift (candidate `synthetic_v2.jsonl` minus baseline):

metric	delta	p	flagged
answer_correctness	-0.107	0.033	yes (*)
citation_grounding	-0.041	0.164	yes
faithfulness	-0.033	0.203	yes
answer_relevance	-0.025	0.410	no
context_precision	0.000	1.000	no
context_recall	0.000	1.000	no

The two `context_*` metrics correctly stayed at 0.000 delta because the candidate fixture changes only the *answers*, not the retrieved contexts; this is what a working drift detector should show. The generation-side metrics moved as expected: `answer_correctness` was the leading indicator ($p < 0.05$), followed by `citation_grounding` and `faithfulness` flagged by delta threshold.

8 7. Ablations

The threshold pair ($\text{delta} = 0.03$, $p = 0.05$) is conservative; in practice you'd tune per-metric. We chose those values because they catch the synthetic-degradation regression without false-positives on a re-run of the same fixture.

9 8. Discussion

The drift-detection framing is the most valuable thing in this repo. Most RAG eval tools score a single run; the *change* in scores across iterations is what actually informs the “should I ship this prompt change” decision. Wrapping a standard t-test around the per-sample scores is two functions of code; not having it is mostly an oversight in the existing tools.

10 9. Limitations

1. **Per-metric tests are independent.** No Bonferroni / BH correction. With 6 metrics at $p < 0.05$, the family-wise false positive rate is $\sim 26\%$.
2. **Heuristic judge is coarse.** It captures topical similarity well but is weak on negation. Use LLM judge for headline numbers.
3. **No bootstrap CIs.** Per-metric means are point estimates.
4. **The fixture is small ($n=10$).** Real RAG eval sets are typically 100-1000 items.

11 10. Future Work

- ☐ Bonferroni / Benjamini-Hochberg correction on drift p-values.
- ☐ Bootstrap CIs on per-metric means.
- ☐ Council judge with reported inter-judge agreement.
- ☐ Failure-category labels per sample (hallucinated / off-topic / under-supported / irrelevant context).
- ☐ Hooks for streaming RAG (chunked answers).

12 11. References

- Es, S., et al. (2023). *RAGAS: Automated Evaluation of Retrieval Augmented Generation*. arXiv:2309.15217.
- Manakul, P., et al. (2023). *SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection*. EMNLP.
- Zheng, L., et al. (2023). *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. NeurIPS.

13 Appendix A. Reproducibility

- Repo: Akshitha024/rag-quality-evaluator, MIT.
- Reproduce: `make eval && cp results/latest__samples.jsonl results/baseline__samples.jsonl && make eval DATA=tests/fixtures/synthetic_v2.jsonl && make drift && make plots.`
- Test artifacts in docs/test_results/.